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(54) **PROCESS LIQUID COMPOSITION FOR LITHOGRAPHY AND PATTERN FORMING METHOD USING SAME**

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(57)

ABSTRACT

Proposed is a process liquid composition for improving a lifting defect level of a photoresist pattern containing a surfactant and for reducing the number of defects of the photoresist pattern, the composition containing a surfactant and having a surface tension of 40 mN/m or less and a contact angle of 60° or smaller in the photoresist pattern having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface in a photoresist pattern process.

5 Claims, No Drawings

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PROCESS LIQUID COMPOSITION FOR LITHOGRAPHY AND PATTERN FORMING METHOD USING SAME

TECHNICAL FIELD

The present invention relates to a process liquid composition for alleviating a lifting defect level of a photoresist pattern and for reducing the number of defects of the photoresist pattern, the photoresist pattern having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface in a photoresist pattern process, and to a method of forming a photoresist pattern using the process liquid composition.

BACKGROUND ART

Generally, a semiconductor is manufactured by a lithographic process in which exposure light is infrared light in a wavelength of 193 nm, 248 nm, 365 nm, or the like. There is intense competition among semiconductor manufacturers for reduction in a critical dimension (hereinafter referred to as a CD).

Accordingly, the finer pattern is to be formed, the narrower wavelength a light source needs to produce. At the present time, a lithographic technology using an extreme ultraviolet (EUV in a wavelength of 13.5 nm) is actively employed. A narrower wavelength may be realized using this lithographic technology.

However, the resistance of EUV photoresist to etching is not yet improved, and thus a photoresist pattern having a high aspect ratio still needs to be used. Accordingly, a pattern lifting defect occurs easily during development, and the number of defects is increased. Consequently, a process margin is greatly reduced in a manufacturing process.

To solve this problem, there is a demand to develop the technology for alleviating a level of a lifting defect that occurs while forming a fine pattern and for reducing the number of defects. The best way to alleviate a pattern lifting defect level and reduce the number of defects may be to improve photoresist performance. However, there is a need to consider a situation where, in practice, it is difficult to develop new photoresist having performance that is satisfactory in terms of all aspects.

There is still a need to develop new photoresist. However, attempts have been made to alleviate the pattern lifting defect level and reduce the number of defects in ways other than addressing this need.

DISCLOSURE

Technical Problem

An objective of the present invention is to develop a process liquid composition for alleviating a level of a pattern lifting defect and reducing the number of defects, the pattern lifting defect occurring after developing photoresist having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface, and to develop a method of forming a photoresist pattern using the process liquid composition.

Technical Solution

Various surfactants are used to manufacture a water-based process liquid composition that is used during a developing process. However, according to the present invention, an

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effective process liquid composition was manufactured using a fluorine-based surfactant and a hydrocarbon-based anionic surfactant.

The use of a hydrocarbon-based non-ionic surfactant with a property like hydrophobicity in manufacturing the water-based process liquid composition in which ultra-pure water is mostly contained may lead to forming a hydrophobic sidewall of a photoresist and thus reducing pattern melting or collapse. However, in this case, the hydrocarbon-based non-ionic surfactants have a strong tendency to agglomerate, resulting in preventing a property of the process liquid composition from being uniform. Theretofore, there is a likelihood that the agglomerating hydrocarbon-based non-ionic surfactants will cause a defect while the process liquid composition is in use. That is, the use of the hydrocarbon-based non-ionic surfactant requires an increase in the usage amount thereof for reducing the pattern melting. Thus, there is a concern that photoresist will be damaged. In addition, the excessive use of an unsuitable surfactant for the purpose of reducing surface tension of the process liquid composition to reduce a capillary force may lead to the pattern melting and rather may further cause the pattern collapse.

In addition, in the case of a hydrocarbon-based cationic surfactant, an active group dissociates into a cation in an aqueous solution, and it is rarely ensured that metal is formed. Thus, there is a concern that a serious defect will be caused to occur in a lithographic process.

According to the present invention, it was verified that the use of the fluorine-based surfactant and the hydrocarbon-based anionic surfactant achieved the noticeable effect of alleviating the pattern lifting defect level and reducing the number of defects. The surface tension and the contact angle, which were much more decreased than in the hydrocarbon-based non-ionic surfactant, increased penetrability and spreadability, leading to contribution to formation of a fine pattern. It was recognized that this contribution resulted in the noticeable effect.

Tetramethylammonium hydroxide is diluted with pure water to a predetermined concentration (2.38% by weight of tetramethylammonium hydroxide is mixed with 97.62% by weight of water for use in most of the photolithographic developing processes) for use as a representative developing solution that is currently used in most of the photolithographic developing processes.

It was verified that a pattern lifting defect was caused in a case where, in a photolithographic process, a photoresist pattern having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface was successively cleaned only with pure water after being developed. Furthermore, it was verified that, in the photolithographic process, the pattern collapse was also caused in a case where a process liquid composition resulting from tetramethylammonium hydroxide being contained in pure water was successively applied after developing or in a case where pure water was successively applied.

It could be estimated that the pattern collapse was caused because the process liquid composition containing tetramethylammonium hydroxide weakened the exposed fine pattern and because the capillary force was great and was non-uniform.

Therefore, in order to prevent the exposed-pattern collapse and to reduce the line width roughness (LWR) and the number of defects additionally required in a process, there is a need to conduct a study on an alkali substance that exerts a relatively weaker force on the exposed pattern than tetramethylammonium hydroxide.

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According to the present invention, it was verified that, in a case where tetraethylammonium hydroxide, tetrapropylammonium hydroxide, and tetrabutylammonium hydroxide was used among alkali substances, not only was the pattern collapse prevented and the LWR, but the number of defects was also reduced.

According to a desirable first embodiment of the present invention, there is provided a process liquid composition for alleviating a level of a lifting defect of a photoresist pattern and for reducing the number of lifting defects of the photoresist pattern, the composition containing a surfactant and having a surface tension of 40 millinewton/meter (mN/m= $\frac{1}{1000}$ newton/meter) or less and a contact angle of 60° or smaller in the photoresist pattern having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface in a photoresist pattern process.

According to a more desirable second embodiment of the present invention, there is provided a process liquid composition for alleviating a level of a lifting defect of a photoresist pattern and for reducing the number of lifting defects of the photoresist pattern, the lifting defect occurring during photoresist developing, the process liquid composition containing: 0.00001 to 0.1% by weight of a fluorine-based surfactant; 0.0001 to 0.1% by weight of a hydrocarbon-based anionic surfactant; 0.0001 to 0.1% by weight of an alkali substance; and 99.7 to 99.99979% by weight of water.

According to the most desirable third embodiment of the present invention, there is a process liquid composition for alleviating a level of a lifting defect of a photoresist pattern and for reducing the number of lifting defects of the photoresist pattern, the lifting defect occurring during photoresist developing, the process liquid composition containing: 0.00001 to 0.1% by weight of a fluorine-based surfactant; 0.001 to 0.1% by weight of a hydrocarbon-based anionic surfactant; 0.001 to 0.1% by weight of an alkali substance; and 99.7 to 99.99979% by weight of water, the composition having a surface tension of 40 mN/m or less and a contact angle of 60° or smaller.

In the process liquid composition according to any one of the first to third embodiments, the fluorine-based surfactant may be selected from the group consisting of fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer, fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, perfluorinated sulfonate, and mixtures thereof.

In the process liquid composition according to any one of the second to third embodiments, wherein the hydrocarbon-based anionic surfactant may be selected from the group consisting of ammonium salt of polycarboxylic acid, sulfonate salt, sulfate ester salt, phosphoric acid ester salt, and mixtures thereof.

In the process liquid composition according to any one of the first to third embodiments, wherein the alkali substance may be selected from the group consisting of tetraethylammonium hydroxide, tetrapropylammonium hydroxide, tetrabutylammonium hydroxide, and mixtures thereof.

According to an aspect of the present invention, there is provided a method of forming a photoresist pattern, the method including: (a) a step of dispensing photoresist on a semiconductor substrate and forming a photoresist film; (b) a step of exposing the photoresist film to light, developing the photoresist film, and forming a photoresist pattern; and (c) a step of cleaning the photoresist pattern with the process liquid composition.

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It was thought that the pattern collapse was caused by the capillary force occurring between patterns when the patterns were cleaned with pure water after developing. However, it was experimentally recognized that only the reduction of the capillary force could neither completely prevent the pattern collapse nor reduce the number of the lifting defects.

The excessive use of the unsuitable surfactant for the purpose of reducing the surface tension of the process liquid composition to reduce the capillary force may lead to the pattern melting and rather may further cause the pattern collapse or increase the number of lifting defects.

In order to alleviate the level of the pattern lifting defect and reduce the number of the pattern lifting defects, it is important to select a surfactant that reduces the surface tension of the process liquid composition and at the same time prevents the melting of the photoresist pattern.

The process liquid composition according to the present invention exerts an enhancing effect on the photoresist and particularly achieves the effect of alleviating the level of the pattern lifting defect and the number of the pattern lifting defects, the pattern lifting defect occurring while developing photoresist having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface.

Advantageous Effects

The process liquid composition according to the present invention achieves the effect of alleviating the level of the pattern lifting defect and the number of the pattern lifting defects, the effect that cannot be achieved only with photoresist when a pattern is formed using the photoresist having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface. The photoresist forming method including the step of cleaning the photoresist pattern with the process liquid composition can achieve the effect of greatly reducing manufacturing cost.

BEST MODE

The present invention will be described in more detail below.

The present invention, which is the result of conducting much research over a long period of time, relates to a "process liquid composition for alleviating a lifting defect level of a photoresist pattern and reducing the number of defects of the photoresist, the process liquid composition containing: 0.00001 to 0.1% by weight of a fluorine-based surfactant selected from the group consisting of fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer, fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, perfluorinated sulfonate, and mixtures thereof; 0.0001 to 0.1% by weight of an anionic surfactant selected from the group consisting of ammonium salt of polycarboxylic acid, sulfonate salt, sulfate ester salt, phosphoric acid ester salt, and mixtures thereof; 0.0001 to 0.1% by weight of an alkali substance selected from the group consisting of tetraethylammonium hydroxide, tetrapropylammonium hydroxide, tetrabutylammonium hydroxide, and mixtures thereof; and 99.7 to 99.99979% by weight of water". Composition components of the process liquid composition according to the present invention and a composition ratio between the components thereof were specified as shown in Embodiments 1 to 60. Composition components and a composition ratio that were in contrast with the above-

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mentioned composition components and composition ratio, respectively, were specified as shown in Comparative Examples 1 to 12.

Desired embodiments of the present invention and comparative examples for comparison therewith will be described below. However, the desired embodiments described below of the present invention are only exemplary, and the present invention is not limited thereto.

MODE FOR INVENTION

Embodiment 1

A process liquid composition for alleviating a collapse level of a photoresist pattern, which contains 0.001% by weight of fluoroacryl carboxylate, 0.01% by weight of ammonium salt of polycarboxylic acid, and 0.005% by weight of tetrabutylammonium hydroxide, was manufactured using the following method.

0.001% by weight of fluoroacryl carboxylate, 0.01% by weight of ammonium salt of polycarboxylic acid, and 0.005% by weight of tetrabutylammonium hydroxide were added into a remaining amount of distilled water and stirred for 5 hours. Then, the resulting liquid was caused to pass through a filter with a size of 0.01 μm to remove fine-sized soluble-solid impurities. In this manner, the process liquid composition for alleviating the collapse level of the photoresist pattern was manufactured.

Embodiments 2 to 60

Process liquid compositions for alleviating a defect level of a photoresist pattern that was the same as a defect level of a photoresist pattern in Embodiment 1 were manufactured according to composition components and component ratios therebetween that were specified as shown in Tables 1 to 12.

Comparative Example 1

Usually, distilled water that was to be used as a cleaning solution in the last process among semiconductor manufacturing processes was prepared.

Comparative Examples 2 to 12

For comparison with embodiments, process liquid compositions were manufactured, as in Embodiment 1, accord-

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ing to the composition components and the component ratios therebetween that were specified as shown in Tables 1 to 12.

Experimental Examples 1 to 60 and Comparative Experimental Examples 1 to 12

Measurements of pattern lifting defect levels and number-of-defects reduction ratios were performed on silicon wafers, patterns on which were formed in Embodiments 1 to 60 and Comparative Examples 1 to 12. The measurements are described as Experimental Examples 1 to 60 and Comparative Experimental Examples 1 to 12. The results of the measurements are shown in Table 13.

(1) Verification of Pattern Lifting Prevention

After exposure energy and focus were split, among a total of 89 blocks, the number of blocks in which a pattern did not collapse was measured using a critical dimension-scanning electron microscope (CD-SEM, manufactured by Hitachi, Ltd).

(2) Number-of-Lifting-Defects Reduction Ratio

Counting of the number A of defects was performed on a photoresist pattern that was rinsed with each process liquid composition sample, using surface defect observation equipment (manufactured by KLA-Tencor Corporation). A value of 100 was assigned to the number B of defects that resulted when the photoresist pattern was rinsed only with pure water. Then, the number A of defects was expressed as a ratio to the number B of defects, that is, as $(A/B) \times 100$.

The number of defects that resulted when rinsing was performed only with pure water was defined as 100. The degree to which the number of defects was decreased (improved) or increased (degraded) when compared with the number of defects resulting from rinsing only with pure water was expressed as a reduction ratio.

(3) Transparency

Transparency of the manufactured process liquid composition was checked with the naked eye and was marked as a transparent or opaque process liquid composition.

(4) Surface Tension and Contact Angle

A surface tension and a contact angle of each of the process liquid compositions were measured using a surface tension measuring instrument [the K100 Force Tensiometer manufactured by KRÜSS GmbH] and a contact angle measuring instrument [the DSA-100 Drop Shape Analyzer manufactured by KRÜSS GmbH].

TABLE 1

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 1	Fluoroacryl carboxylate	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 2	Fluoroalkyl ether	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 3	Fluoroalkylene ether	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 4	Fluoroalkyl sulfate	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 5	Fluoroalkyl phosphate	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 6	Fluoroacryl co-polymer	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 7	Fluoro co-polymer	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 8	Perfluorinated acid	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840

TABLE 1-continued

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 9	Perfluorinated carboxylate	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 10	Perfluorinated sulfonate	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Comparative Example 1	—	—	—	—	—	—	Distilled water	100
Comparative Example 2	—	—	—	—	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9950

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TABLE 2

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 11	Fluoroacryl carboxylate	0.001	Sulfonate salt	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 12	Fluoroalkyl ether	0.001	Sulfonate salt	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 13	Fluoroalkylene ether	0.001	Sulfonate salt	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 14	Fluoroalkyl sulfate	0.001	Sulfonate salt	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 15	Fluoroalkyl phosphate	0.001	Sulfonate salt	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 16	Fluoroacryl co-polymer	0.001	Sulfonate salt	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 17	Fluoro co-polymer	0.001	Sulfonate salt	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 18	Perfluorinated acid	0.001	Sulfonate salt	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 19	Perfluorinated carboxylate	0.001	Sulfonate salt	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 20	Perfluorinated sulfonate	0.001	Sulfonate salt	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840

TABLE 3

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 21	Fluoroacryl carboxylate	0.00001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.98499
Embodiment 22	Fluoroacryl carboxylate	0.0001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9849
Embodiment 1	Fluoroacryl carboxylate	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 23	Fluoroacryl carboxylate	0.01	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9750
Embodiment 24	Fluoroacryl carboxylate	0.1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.8850
Comparative Example 3	Fluoroacryl carboxylate	1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	98.9850

TABLE 4

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 25	Fluoroalkyl ether	0.00001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.98499
Embodiment 26	Fluoroalkyl ether	0.0001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9849
Embodiment 2	Fluoroalkyl ether	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 27	Fluoroalkyl ether	0.01	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9750
Embodiment 28	Fluoroalkyl ether	0.1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.8850
Comparative Example 4	Fluoroalkyl ether	1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	98.9850

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TABLE 5

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 29	Fluoroalkylene ether	0.00001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.98499
Embodiment 30	Fluoroalkylene ether	0.0001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9849
Embodiment 3	Fluoroalkylene ether	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 31	Fluoroalkylene ether	0.01	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9750
Embodiment 32	Fluoroalkylene ether	0.1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.8850
Comparative Example 5	Fluoroalkylene ether	1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	98.9850

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TABLE 6

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodimen 33	Fluoroalkyl sulfate	0.00001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.98499
Embodiment 34	Fluoroalkyl sulfate	0.0001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9849
Embodiment 4	Fluoroalkyl sulfate	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 35	Fluoroalkyl sulfate	0.01	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9750
Embodiment 36	Fluoroalkyl sulfate	0.1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.8850
Comparative Example 6	Fluoroalkyl sulfate	1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	98.9850

TABLE 7

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 37	Fluoroalkyl phosphate	0.00001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.98499
Embodiment 38	Fluoroalkyl phosphate	0.0001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9849
Embodiment 5	Fluoroalkyl phosphate	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 39	Fluoroalkyl phosphate	0.01	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9750
Embodiment 40	Fluoroalkyl phosphate	0.1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.8850
Comparative Example 7	Fluoroalkyl phosphate	1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	98.9850

TABLE 8

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 41	Fluoroacryl co-polymer	0.00001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.98499
Embodiment 42	Fluoroacryl co-polymer	0.0001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9849
Embodiment 6	Fluoroacryl co-polymer	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 43	Fluoroacryl co-polymer	0.01	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9750
Embodiment 44	Fluoroacryl co-polymer	0.1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.8850
Comparative Example 8	Fluoroacryl co-polymer	1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	98.9850

TABLE 9

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 45	Fluoro co-polymer	0.00001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.98499
Embodiment 46	Fluoro co-polymer	0.0001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9849
Embodiment 7	Fluoro co-polymer	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 47	Fluoro co-polymer	0.01	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9750
Embodiment 48	Fluoro co-polymer	0.1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.8850
Comparative Example 9	Fluoro co-polymer	1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	98.9850

TABLE 10

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 49	Perfluorinated acid	0.00001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.98499
Embodiment 50	Perfluorinated acid	0.0001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9849
Embodiment 8	Perfluorinated acid	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 51	Perfluorinated acid	0.01	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9750
Embodiment 52	Perfluorinated acid	0.1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.8850
Comparative Example 10	Perfluorinated acid	1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	98.9850

TABLE 11

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 53	Perfluorinated carboxylate	0.00001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.98499
Embodiment 54	Perfluorinated carboxylate	0.0001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9849
Embodiment 9	Perfluorinated carboxylate	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 55	Perfluorinated carboxylate	0.01	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9750
Embodiment 56	Perfluorinated carboxylate	0.1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.8850
Comparative Example 11	Perfluorinated carboxylate	1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	98.9850

TABLE 12

	Surfactant		Surfactant		Alkali substance		Distilled water	
	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)	Name	Content (% by weight)
Embodiment 57	Perfluorinated sulfonate	0.00001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.98499
Embodiment 58	Perfluorinated sulfonate	0.0001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9849
Embodiment 10	Perfluorinated sulfonate	0.001	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9840
Embodiment 59	Perfluorinated sulfonate	0.01	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.9750
Embodiment 60	Perfluorinated sulfonate	0.1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	99.8850
Comparative Example 12	Perfluorinated sulfonate	1	Ammonium salt of polycarboxylic acid	0.01	Tetrabutylammonium hydroxide	0.005	Distilled water	98.9850

Experimental Examples 1 to 60 and Comparative Experimental Examples 1 to 12

Measurements of the pattern lifting defect level, the number-of-defects reduction ratio, the transparency, the contact angle, and the surface tension were performed on the silicon wafers, the patterns on which were formed in Embodiments 1 to 60 and Comparative Examples 1 to 12. The measurements are described as Experimental Examples

60 1 to 60 and Comparative Experimental Examples 1 to 12. The results of the measurements are shown in Table 13.

(1) Verification of Pattern Lifting Prevention

65 After exposure energy and focus were split, among a total of 89 blocks, the number of blocks in which a pattern did not collapse was measured using the critical dimension-scanning electron microscope (CD-SEM, manufactured by Hitachi, Ltd).

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(2) Number of Lifting Defects

Counting of the number A of defects was performed on a photoresist pattern that was rinsed with each process liquid composition sample, using the surface defect observation equipment (manufactured by KLA-Tencor Corporation). A value of 100 was assigned to the number B of defects that resulted when the photoresist pattern was rinsed only with pure water. Then, the number A of defects was expressed as a ratio to the number B of defects, that is, as $(A/B) \times 100$.

(3) Transparency

Transparency of the manufactured process liquid composition was checked with the naked eye and was marked as a transparent or opaque process liquid composition.

(4) Contact Angle and Surface Tension

A surface tension and a contact angle of each of the process liquid compositions were measured using the contact angle measuring instrument [the DSA-100 Drop Shape Analyzer manufactured by KRÜSS GmbH] and the surface tension measuring instrument [the K100 Force Tensiometer manufactured by KRÜSS GmbH].

TABLE 13

	The number of blocks with no lifting defects	Number- of-defects reduction ratio (%)	Trans- parency	Contact angle (°)	Surface tension (mN/m)
Experimental Example 1	80	25	transparent	49	22
Experimental Example 2	78	30	transparent	55	25
Experimental Example 3	78	35	transparent	52	22
Experimental Example 4	77	35	transparent	57	23
Experimental Example 5	77	40	transparent	56	23
Experimental Example 6	75	55	transparent	57	26
Experimental Example 7	72	62	transparent	58	30
Experimental Example 8	73	60	transparent	52	29
Experimental Example 9	76	55	transparent	54	27
Experimental Example 10	75	48	transparent	53	25
Experimental Example 11	77	30	transparent	50	22
Experimental Example 12	78	38	transparent	55	28
Experimental Example 13	78	42	transparent	53	24
Experimental Example 14	77	44	transparent	58	27
Experimental Example 15	76	58	transparent	56	26
Experimental Example 16	76	64	transparent	58	30
Experimental Example 17	71	74	transparent	59	33
Experimental Example 18	72	59	transparent	53	32
Experimental Example 19	74	61	transparent	54	29
Experimental Example 20	74	52	transparent	54	28
Experimental Example 21	64	44	transparent	60	39
Experimental Example 22	70	38	transparent	55	29
Experimental Example 23	77	28	transparent	42	19
Experimental Example 24	76	30	transparent	36	15

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TABLE 13-continued

	The number of blocks with no lifting defects	Number- of-defects reduction ratio (%)	Trans- parency	Contact angle (°)	Surface tension (mN/m)
Experimental Example 25	58	50	transparent	59	40
Experimental Example 26	65	36	transparent	57	33
Experimental Example 27	75	32	transparent	52	22
Experimental Example 28	74	33	transparent	49	19
Experimental Example 29	60	55	transparent	59	37
Experimental Example 30	68	42	transparent	56	28
Experimental Example 31	75	38	transparent	49	19
Experimental Example 32	73	39	transparent	45	16
Experimental Example 33	61	52	transparent	60	36
Experimental Example 34	67	44	transparent	58	30
Experimental Example 35	74	37	transparent	54	21
Experimental Example 36	73	38	transparent	50	17
Experimental Example 37	60	60	transparent	59	37
Experimental Example 38	65	49	transparent	57	31
Experimental Example 39	73	42	transparent	54	20
Experimental Example 40	73	44	transparent	52	17
Experimental Example 41	59	77	transparent	60	38
Experimental Example 42	67	65	transparent	58	32
Experimental Example 43	71	58	transparent	55	23
Experimental Example 44	70	60	transparent	53	20
Experimental Example 45	52	80	transparent	60	40
Experimental Example 46	60	69	transparent	60	34
Experimental Example 47	68	64	transparent	56	27
Experimental Example 48	66	65	transparent	55	24
Experimental Example 49	61	82	transparent	57	38
Experimental Example 50	69	73	transparent	55	35
Experimental Example 51	72	62	transparent	50	24
Experimental Example 52	71	65	transparent	48	20
Experimental Example 53	63	68	transparent	58	39
Experimental Example 54	70	62	transparent	56	33
Experimental Example 55	74	55	transparent	51	23
Experimental Example 56	73	56	transparent	50	22
Experimental Example 57	61	57	transparent	60	38
Experimental Example 58	68	50	transparent	55	29
Experimental Example 59	72	49	transparent	50	20
Experimental Example 60	71	50	transparent	48	18

TABLE 13-continued

	The number of blocks with no lifting defects	Number- of-defects reduction ratio (%)	Trans- parency	Contact angle (°)	Surface tension (mN/m)
Comparative Experimental Example 1	46	100	transparent	89	70
Comparative Experimental Example 2	40	95	transparent	58	67
Comparative Experimental Example 3	58	150	transparent	35	14
Comparative Experimental Example 4	54	172	transparent	50	19
Comparative Experimental Example 5	52	184	transparent	44	16
Comparative Experimental Example 6	51	186	opaque	47	16
Comparative Experimental Example 7	50	180	opaque	49	16
Comparative Experimental Example 8	51	210	opaque	53	20
Comparative Experimental Example 9	50	235	opaque	54	22
Comparative Experimental Example 10	50	170	opaque	46	38
Comparative Experimental Example 11	52	168	opaque	49	21
Comparative Experimental Example 12	51	174	opaque	47	18

From the comparison of Experimental examples 1 to 60 with Comparative Experimental examples 1 to 12, it could be seen that, when the number of blocks in which a pattern did not collapse was 50 or greater and the number-of-defects reduction ratio was 90% or less, a more improved result were obtained than in Comparative Experimental Example 1.

It could be seen that the pattern lifting defect level was much more alleviated and the number of defects was much more reduced in Experimental Examples 1 to 60 than in Comparative Experimental Examples 1 to 12. The process liquid composition that was used in Experimental Examples 1 to 60 contained: 0.00001 to 0.1% by weight of a fluorine-based surfactant selected from among fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer, fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, and perfluorinated sulfonate; 0.0001 to 0.1% by weight of an anionic surfactant selected from among polycarboxylate salt, sulfonate salt, sulfate ester salt, and phosphoric acid ester salt; 0.0001 to 0.1% by weight of an alkali substance selected from among tetraethylammonium hydroxide, tetrapropylammonium hydroxide, and tetrabutylammonium hydroxide; and 99.7 to 99.99979% by weight of water.

In addition, it could be seen that, desirably, effects of alleviating the pattern lifting defect level and reducing the number of defects was much more increased in the experimental examples 1 to 60 than in Comparative Experimental Examples 1 to 12. The process liquid composition that was used in Experimental Examples 1 to 60 contained: 0.0001 to

0.1% by weight of a fluorine-based surfactant selected from among fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer, fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, and perfluorinated sulfonate; 0.001 to 0.1% by weight of a hydrocarbon-based anionic surfactant selected from among polycarboxylate salt, sulfonate salt, sulfate ester salt, and phosphoric acid ester salt; 0.001 to 0.1% by weight of an alkali substance selected from among tetraethylammonium hydroxide, tetrapropylammonium hydroxide, and tetrabutylammonium hydroxide; and 99.7 to 99.9979% by weight of water.

In addition, it could be seen that, more desirably, effects of alleviating the pattern lifting defect level and reducing the number of defects was much more increased in the experimental examples 1 to 60 than in Comparative Experimental Examples 1 to 12. The process liquid composition that was used in Experimental Examples 1 to 60 contained: 0.001 to 0.1% by weight of a fluorine-based surfactant selected from among fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer, fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, and perfluorinated sulfonate; 0.01 to 0.1% by weight of a hydrocarbon-based anionic surfactant selected from among polycarboxylate salt, sulfonate salt, sulfate ester salt, and phosphoric acid ester salt; 0.01 to 0.1% by weight of an alkali substance selected from among tetraethylammonium hydroxide, tetrapropylammonium hydroxide, and tetrabutylammonium hydroxide; and 99.7 to 99.979% by weight of water.

The result of measuring the collapse level of the photoresist pattern in Embodiment 1 for evaluation, was that the number of blocks in which the pattern did not collapse was 80.

The result of measuring the collapse level of the photoresist pattern in Comparative Experimental Example 1 for evaluation, was that the number of blocks in which the pattern did not collapse was 46.

The specific aspects of the present invention are described in detail above. It would be apparent to a person of ordinary skill in the art to which the present invention pertains that this specific description is only for the desired embodiments and do not impose any limitation on the scope of the present invention. Therefore, a substantial scope and a scope equivalent thereto must be defined by the following claims.

The invention claimed is:

1. A process liquid composition for alleviating a lifting defect level of a photoresist pattern and for reducing the number of defects of the photoresist pattern, the composition containing a surfactant, and having a surface tension of 40 mN/m or less and a contact angle of 60° or smaller in the photoresist pattern having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface in a photoresist pattern process, the composition comprising:

0.00001 to 0.1% by weight of a fluorine-based surfactant;
0.0001 to 0.1% by weight of a hydrocarbon-based anionic surfactant;

0.0001 to 0.1% by weight of an alkali substance that is selected from a group consisting of tetraethylammonium hydroxide, tetrapropylammonium hydroxide, tetrabutylammonium hydroxide, and mixtures thereof; and

a remaining percentage by weight of water.

2. The process liquid composition of claim 1, comprising:
0.0001 to 0.1% by weight of the fluorine-based surfactant;
0.001 to 0.1% by weight of the hydrocarbon-based
anionic surfactant;
0.001 to 0.1% by weight of the alkali substance; and 5
99.7 to 99.9979% by weight of the water.

3. The process liquid composition of claim 2, wherein the
fluorine-based surfactant is selected from the group consist-
ing of fluoroacryl carboxylate, fluoroalkyl ether, fluoroal-
kylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, 10
fluoroacryl co-polymer, fluoro co-polymer, perfluorinated
acid, perfluorinated carboxylate, perfluorinated sulfonate,
and mixtures thereof.

4. The process liquid composition of claim 2, wherein the
hydrocarbon-based anionic surfactant is selected from the 15
group consisting of ammonium salt of polycarboxylic acid,
sulfonate salt, sulfate ester salt, phosphoric acid ester salt,
and mixtures thereof.

5. A method of forming a photoresist pattern, the method
comprising: 20

- (a) dispensing photoresist on a semiconductor substrate
and forming a photoresist film;
- (b) exposing the photoresist film to light, and developing
the photoresist film to form a photoresist pattern; and
- (c) cleaning the photoresist pattern with the process liquid 25
composition of claim 1.

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